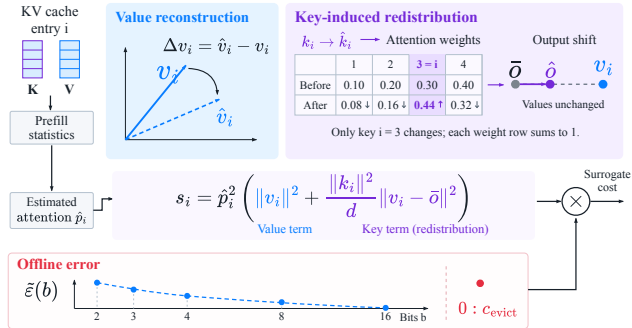


(a) Output-distortion surrogate



(b) Shared-budget allocation

Per-entry bit selection

$$b_i(\lambda) = \arg \min_{b \in \{0, 2, 3, 4, 8, 16\}} [s_i \tilde{\epsilon}(b) + \lambda b]$$

Distortion term Bit cost

λ bisection · parallel choices · discrete repair

Same average budget: 2 bits

Entry i	1	2	3	4	5	6	7	8
Uniform	2	2	2	2	2	2	2	2
Keep / drop	0	0	0	0	16	0	0	0
ODM-KV	0	2	0	3	8	0	3	0

Shared across KV heads and tokens within each layer

(c) Packed KV cache

2-bit bank

$i = 2$
K V

3-bit bank

$i = 4$	$i = 7$
K V	K V

8-bit bank

$i = 5$
K V

One tile = one KV entry (schematic)
0-bit entries are evicted

BF16 tail

Recent KV entries

new	new	new
K V	K V	K V

Current $q_t \rightarrow \Sigma \rightarrow \hat{o}_t$

Fused attention